Reliable Nuclei Segmentation Across Tissues and Cell Classes

A targeted study of missed Dead nuclei
and tissue-specific performance on PanNuke

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Project 5

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Project summary

Can one model detect, separate, and classify nuclei reliably across the tissues and cell classes represented in PanNuke? We reproduce strong baselines, analyze where they fail, and test a targeted improvement. Dead nuclei are our main case because they are rare and frequently missed. Their improvement will be evaluated alongside the other four classes and within tissues with sufficient label support; a higher overall Dead score alone cannot establish broad reliability.

Our completed studies show persistent tissue differences and a substantial Dead detection deficit. Increasing image scale recovers some nuclei at high computational cost. A controlled change to supervision for cropped nuclei fails its predefined quality and mechanism criteria. We next propose PM-Net, which uses candidate points to predict individual masks and class labels directly. Matched comparisons will test whether this representation improves detection while preserving overall quality and limiting inference latency to twice the baseline.

Intellectual merit

The contribution is a controlled study of a specific failure: nuclei that remain hard to detect despite strong pretrained image features. We separate finding an object, drawing its boundary, and assigning its class. Tissue-by-class analysis will establish where any improvement holds and where it fails. The official PanNuke protocol evaluates new images from represented tissues; transfer to excluded tissue types is a separate, optional question.

Broader impacts

Missed nuclei and incorrect class labels can bias estimates of tissue composition. Reproducible evaluation, explicit subgroup coverage, and computational-cost reporting will help researchers judge whether a segmentation improvement is useful beyond its average score.

1. Research question, scope, and aims

Problem and motivation

Digital pathology requires a separate mask and class label for each nucleus in an H&E image. Touching nuclei must remain separate, and small or damaged nuclei must not disappear. Tissue appearance, cell density, nuclear morphology, and staining vary. A model can therefore achieve a reasonable average score while performing poorly on particular tissues or classes.

Project 5 asks for a reproduced baseline, evaluation by tissue and nucleus class, and one targeted improvement for difficult tissues or rare classes. We interpret consistency as dependable performance across the supported tissue-class groups, rather than identical scores for groups of different difficulty.

Two questions with different evidence requirements

Question	Evidence needed	Role in this project
Reliability across represented tissues and classes	Official splits; per-tissue and per-class scores; sample counts and error analysis.	Primary scope. Dead detection is a targeted case within five-class prediction.
Transfer to an excluded tissue type	Exclude target tissue from task training and tuning; compare with a matched target-present reference.	Optional extension under a separate frozen protocol.

PanNuke contains 7,901 prepared H&E patches with 189,744 labeled nuclei in five classes across 19 tissues [1,11]. These coarse labels define a nucleus-analysis task, not comprehensive pathology understanding. Strong pretrained features may help, but their value for typing does not guarantee that a small nucleus will be found and segmented. We therefore test a detection hypothesis before adding further semantic machinery.

Specific aims

Aim 1: locate the reliability gaps. Reproduce HoVer-Net and CellViT-UNI, then analyze missed, merged, split, and mistyped nuclei by tissue and class. Completed subgroup results motivate the intervention and limit its claims.

Aim 2: test crop-related supervision. Determine whether retaining geometric targets for newly truncated small nuclei improves detection. This controlled study is complete and fails its predefined criteria.

Aim 3: test direct instance prediction. Compare point-conditioned masks with the CellViT-UNI decoder under matched training conditions. Require better Dead detection and PQ, preserved five-class quality, and acceptable inference cost.

Dead is a dataset label that includes apoptotic and necrotic nuclei. Improving this class addresses a rare-class weakness; it does not demonstrate recognition of unseen cell classes or unseen tissues.

2. Prior work and preliminary evidence

Baseline and proposed difference

HoVer-Net predicts nucleus pixels, horizontal/vertical (HV) position maps, and types, then separates instances with watershed [2]. CellViT adds transformer features while retaining this decoding route [3]. PM-Net will instead predict masks from point-conditioned candidates, drawing on point prompts [4], set prediction [5], and spatial adaptation [6]. We retain a matched UNI encoder [10] so that decoder changes can be evaluated without changing the pretrained representation.

Tissue differences persist within a cell class

Our original three-split reproductions reach mPQ/bPQ 0.4564/0.6613 for HoVer-Net and 0.4995/0.6653 for CellViT-UNI. A descriptive audit of the later x1+M/S baseline uses nine frozen split/seed pairs [11]. The table averages three seeds within each split, then the three splits.

Tissue	mPQ	Connective PQ	Unique test images
Lung	0.4064	0.2542	131
Skin	0.4264	0.3011	100
Colon	0.4370	0.3967	972
Breast	0.5155	0.4398	1,602

Lung, Skin, and Colon rank in the bottom five in all three split-level seed means. The connective groups above each have at least 25 positive images per underlying test fold. Comparing the same class reduces the influence of different class mixtures, but does not control size, morphology, staining, density, or annotation. These results establish uneven performance on represented tissues, without identifying its cause.

Why focus on Dead detection?

Across all three unique folds, 2,908 Dead nuclei occur in 232 positive images; 59.5% are in Lung, and seven tissues have no Dead labels. In the matched baseline, 49.1% of Dead labels match a prediction at the evaluator's class-agnostic IoU criterion; 74.5% of matched Dead have the correct type. This supports detection as a substantial deficit relative to the supplied labels, while leaving mask quality and typing relevant.

The frozen EF-P2 scale-addition method changes Dead PQ by +0.0208 in Lung, +0.0093 in Colon, and -0.0014 in Stomach. Its overall gain is +0.0070 at about 6.1 times inference cost [7,11]. This motivates a more economical detection study; it does not establish a broadly better model or uniquely justify PM-Net.

This audit reuses historical test predictions and is exploratory, not independent confirmation. Splits 1 and 2 share test fold 3; unique test counts include folds 1 and 3 once. The full-data support counts describe label coverage. New method choices will use training and validation data only.

3. Completed study: crop-related supervision

Spatial cropping can leave only part of a nucleus visible. The existing target generator sets HV values to zero for instances smaller than 30 working pixels, while foreground and type labels remain available. Aim 2 tests whether retaining HV targets for newly truncated small instances improves detection. A prior crop probe motivates the question but does not establish zero HV targets as the cause of misses [8].

Arm	Crop probability	HV target rule
A0: retrained baseline	0.1	Minimum area 30
A1: crop-only control	0.5	Minimum area 30
A2: selective supervision	0.5	Minimum 1 for newly truncated instances with visible area below 30
A3: small-object control	0.5	Minimum 1 for all instances; targets differ only below area 30

All arms use seeds 19 and 1, 130 epochs, matched initialization and optimization, and final-checkpoint evaluation at native resolution without test-time augmentation. A1-A3 share the augmentation random sequence. Instance identities before and after cropping identify newly truncated objects; areas are measured after geometric transformations. A2 versus A1 isolates selective HV supervision from stronger cropping [8].

Original-image validation: quality criteria fail

All eight runs and fold-2 evaluations are complete. Relative to A0, A2 increases official Dead PQ by 0.0222 and 0.0269 for seeds 19 and 1. However, mPQ falls by 0.0091 and 0.0034, and bPQ by 0.0062 and 0.0046. Both seeds exceed the predefined 0.002 quality-loss tolerance; seed 1 also loses 0.0044 strict Dead PQ. Against the crop-only A1 control, A2 loses 0.0131 official Dead PQ and 0.0098 strict Dead PQ on average. The Dead increase over A0 cannot be credited to selective HV supervision.

Fixed-window analysis: the proposed rescue is not supported

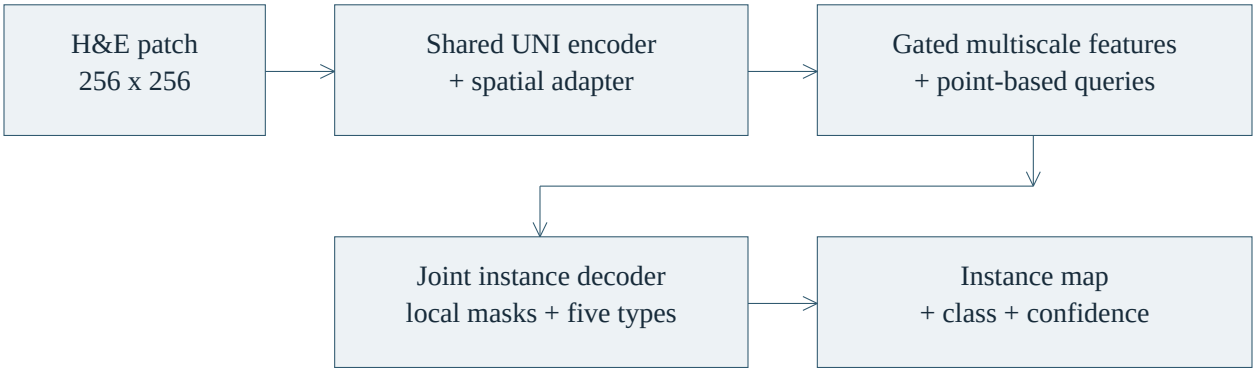
Among 116 eligible Dead objects in 38 validation images, A1 and A2 miss 51 and 54 objects for seed 19, and 52 and 51 for seed 1. Mean A1-minus-A2 miss-rate improvement is -0.0086, with paired 95% interval [-0.0409, +0.0177]. This is below the required +0.03 and inconsistent across seeds. Inference covers all 2,523 fixed windows; 77 eligible objects are from Lung, and five only touch the crop edge without losing pixels. Tissue-stratified image bootstrap conditions on the two trained seeds and eligible images [8].

Implication for the proposed model

The tested supervision recipe fails both its quality and mechanism criteria and will not be expanded. This result does not reject all visibility-aware approaches or prove that watershed causes the remaining errors. Aim 3 asks a separate question: can direct, point-conditioned mask prediction improve detection and instance separation under matched conditions?

4. Proposed study: PM-Net direct instance prediction

PM-Net will predict one mask and class distribution for each candidate nucleus [9]. The hypothesis is that explicit interior-point supervision and individual mask prediction can improve localization and separation of difficult nuclei. This is a testable design choice, not an established solution to tissue variability or a demonstrated novelty claim over existing query-based models.



Proposed PM-Net. One shared image encoding; no HV maps or watershed.

Computational steps

1. Encode the image once. UNI provides shared image features. A compact CNN adapter supplies finer spatial detail; learned gates combine the scales. The initial prototype uses native 256 x 256 patches and the same pretrained encoder initialization as CellViT-UNI [10].

2. Propose interior points. A directly supervised heatmap predicts candidate locations. The proposed target is the visible mask pixel farthest from its boundary, with deterministic tie-breaking. Each selected point initializes a query: a feature vector representing a candidate nucleus. Audit disconnected masks, empty images, and augmented instance counts before fixing the rule.

3. Predict masks and types. Queries read shared local and contextual features to predict masks, object confidence, and five-class probabilities. One-to-one matching associates candidates with labels using location, class, and mask agreement. Training combines point focal loss, mask cross-entropy and Dice losses, and type cross-entropy; weights are chosen on validation data. Direct heatmap supervision trains the point head despite discrete point selection.

4. Resolve overlaps. Filter low-confidence candidates, then use confidence-weighted mask probabilities and a background threshold to assign pixels. Freeze these rules on validation data before confirmation. Removing HV and watershed still leaves explicit decoding choices; minimum-area filtering must be checked for losses of small nuclei.

Ablations and feasibility

Compare a basic query decoder with point initialization and gated feature fusion added separately. Include an ordinary adapter with matched parameter count to test whether extra capacity explains the gain. Keep five-class prediction throughout; additional truncation supervision and auxiliary type grouping are later options, not prerequisites.

A training-only audit found up to 276 instances per original patch, so 256 candidates cannot represent every label. We will profile 512 candidates, verify augmented counts, and measure the complete attention, mask-output, matching, and decoding costs before training. Sparse feature reads may reduce attention cost but do not remove the other costs.

5. Experimental design and evaluation

Development, followed by conditional confirmation

Development uses training fold 1, validation fold 2, paired seeds 19 and 1, 130 epochs, and encoder unfreezing at epoch 25. Match UNI initialization and the training recipe across PM-Net and CellViT-UNI; record parameter counts and compute for each ablation. Fix architecture, losses, augmentation, decoding, and decision criteria before confirmation.

If development criteria and resources permit, evaluate final checkpoints under train/validation/test folds 1/2/3, 2/1/3, and 3/2/1. Report each split and seed before averaging. Splits 1 and 2 share test fold 3. Disclose historical test reuse and the lack of identifiers needed to verify patient-level separation. These official splits assess new images from represented tissues, not transfer to excluded tissues.

Measure the failure and its coverage

Report official type-aware mPQ, class-agnostic bPQ, and per-class PQ. Separate missed instances, low-overlap masks, merges, splits, and matched-instance typing errors. Dead recall and precision will show whether better PQ comes from finding more nuclei or from other changes. Strict mPQ and strict Dead PQ retain absent-class false-positive penalties.

A 19-tissue by five-class table will show baseline quality, paired changes, and image/nucleus counts. Compare the same class across tissues; per-tissue mPQ also reflects class composition. Report low-performing tissues and their uncertainty. Unsupported cells are unavailable, and sparse cells cannot establish preserved performance. A pooled Dead gain will receive a tissue-limited interpretation if support or benefits are concentrated.

Use paired, tissue-stratified image bootstrap intervals for matched comparisons. Resample windows from one source patch together, and share fold-3 draws between overlapping splits. Report each seed separately: intervals condition on the trained models and do not estimate patient-level or population-of-seeds generalization. Subgroup analyses describe coverage; they do not create new selection rules after test inspection.

Proposed PM-Net development criteria

Before training, fix a mean interior Dead recall gain of at least 0.03 with a positive paired 95% lower confidence bound. Mean Dead PQ must improve by at least 0.005; recall and Dead PQ changes must be positive in each seed. Per-seed mPQ/bPQ losses are limited to 0.002. Strict mPQ and strict Dead PQ must not decline, and mean merge/split rates must not increase. These proposed criteria are separate from the completed, frozen supervision study.

Complete inference, including decoding, must take no more than twice the matched single-scale baseline latency. Repeat measurements on the same hardware and batch settings, reporting peak memory and shared-device variability. Only a prototype meeting quality and cost criteria proceeds to confirmation as resources permit.

6. Milestones, feasibility, and deliverables

Schedule	Milestone
Week 1	Consolidate completed baselines, tissue/class analysis, and supervision results.
Week 2	Finalize PM-Net specification; verify target construction, matching, capacity, and runtime.
Weeks 3-4	Run the two-seed development comparison and core ablations.
Weeks 5-6	Complete the report; pursue confirmation if quality, cost, and resources permit.

The six-week schedule begins at proposal agreement. A narrow, tested result takes priority over adding more architectures or objectives.

Scope and expected outputs

Minimum result. Reproduced baselines, a tested targeted supervision modification with its negative result, and tissue-by-class error tables with counts and uncertainty. This addresses the required baseline, analysis, and method-development tasks without claiming a performance gain.

Target result. PM-Net code, a feasible prototype, trained development models, and two-seed ablations showing whether direct instance prediction improves Dead detection within the quality and latency criteria. A well-controlled negative result is useful if it explains the remaining limitations.

Stretch goals. Cross-split confirmation and optional external evaluation with explicit class mapping and overlap exclusion. A separate tissue-holdout study would exclude selected tissues from task training and tuning and compare with a matched target-present reference. This would estimate the cost of task-level exclusion, not universal tissue generalization. Pathology pretraining may already include the held-out tissue types.

Main risks and mitigation

Sparse or uncertain labels. Dead labels are concentrated in few tissues, and expert review is unavailable. Preserve the supplied labels, report support and uncertainty, and avoid biological claims from detector agreement. Limit conclusions to groups with usable evidence.

Localization does not guarantee good masks. A valid point may still yield poor overlap, duplicates, or merged objects. Separate these errors so that a recall change is not mistaken for complete instance improvement.

Compute and implementation cost. Profile the full prototype before training and retain a simple matched baseline. Estimated compute is 2-4 A100 GPU-days for development and a further 4-8 for confirmation, subject to allocation. Check free memory and storage before scheduling. Stop expansion if development criteria fail; document the failure rather than relaxing the criteria.

Deliverables and broader impact

Deliver versioned code and evaluation settings, model/data provenance, quantitative tables, and a final report with annotated failure cases. These outputs will show which detection changes help, where the evidence is limited, and what they cost. Tissue-specific reporting makes the implications for cell-composition measurements clearer than an aggregate score alone.

References and project evidence

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[7] BDD26 Project Team. (2026). PanNuke nuclei segmentation: research plan, experiment findings, and stage reports I-V. Internal project documentation, Johns Hopkins University.

Versioned project documentation

[8] BDD26 Project Team. (2026). Visibility-supervision experiment: frozen specification, completed development evaluations, and fixed-window mechanism analysis. Internal research protocol and experiment records, Johns Hopkins University.

Frozen supervision specification

Completed development results | Fixed-window mechanism analysis

[9] BDD26 Project Team. (2026). Next-stage research proposal: a new non-HoVer-Net model route. Internal design proposal, Johns Hopkins University.

Versioned new-model concept

[10] Chen, R. J., et al. (2024). Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30, 850-862.

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[11] BDD26 Project Team. (2026). Tissue and class consistency audit of frozen PanNuke evaluations. Nine matched split/seed pairs, unique-fold support counts, and exploratory paired image bootstrap.

Audit methods, results, and reproducible sources

Team contributions to this proposal

This page is separate from the main proposal.

The team's proposal contributions are summarized below.

Zhiyu Song

Helped shape the research question around tissue- and class-specific segmentation reliability, and contributed to the project's biological motivation and scope.

Yuchen Ai

Contributed to literature review and method comparisons, helping frame the baseline choices and the proposed point-conditioned instance-segmentation approach.

Xinhao Jin

Implemented and maintained core segmentation and evaluation code, analyzed tissue-by-class results and failure modes, and integrated findings into the proposal.

Zheng Jian

Contributed to experimental-design ideas, validation and metric choices, and interpretation of results; reviewed the PM-Net ablations and feasibility criteria.